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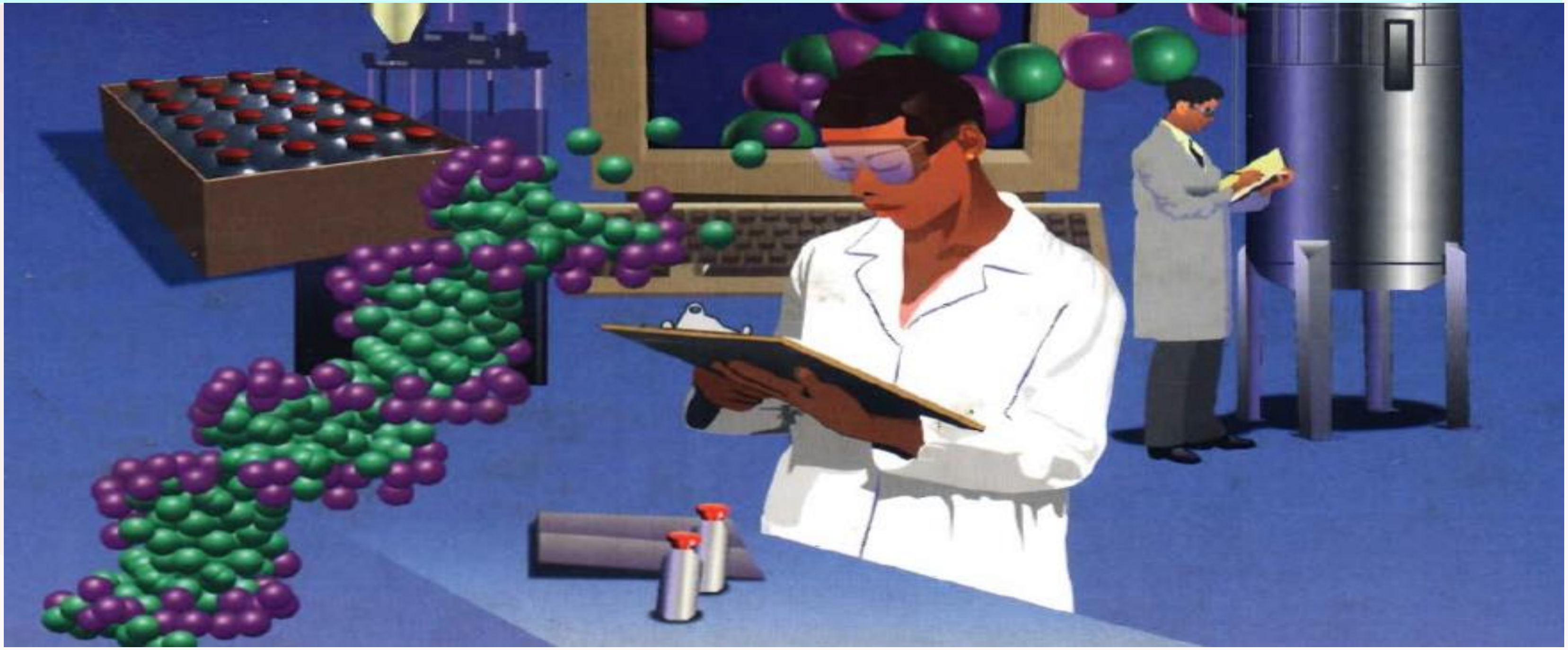
现代分析测试技术
Modern Instrumental Analysis



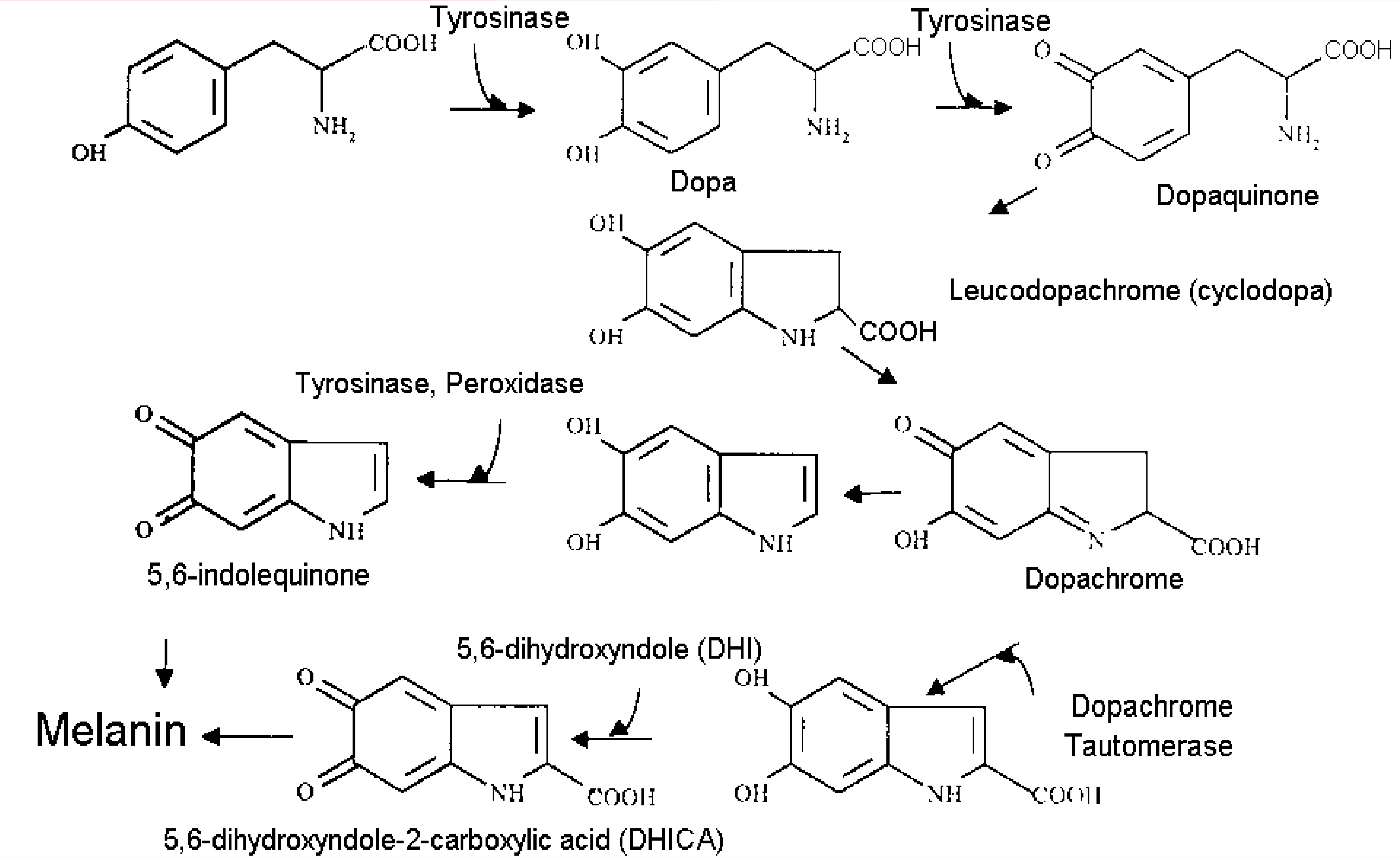
紫外-可见吸收光谱法

UV-visible molecular absorption spectroscopy

大海为什么是蓝色的？？？

防晒霜是如何挡住紫外线的？？

为什么肤白容易晒伤？？？


Tyrosinase
COOH
NH2
OH
Tyrosinase
OH
OH
COOH
NH2
Dopa
O
O
COOH
NH2
Dopaquinone
OH
OH
NH
COOH
Leucodopachrome (cyclodopa)
Tyrosinase, Peroxidase
O
O
NH
5,6-indolequinone
OH
OH
NH
O
OH
N
COOH
Dopachrome
5,6-dihydroxyndole (DHI)
Melanin
O
O
NH
COOH
OH
OH
NH
COOH
Dopachrome
Tautomerase
5,6-dihydroxyndole-2-carboxylic acid (DHICA)

Definitions and Units

- Radiation is a form of energy and we are constantly reminded of its presence via our sense of sight and ability to feel radiant heat. It may be considered in terms of a wave motion where the wavelength, λ, is the distance between two successive peaks. The frequency, ν, is the number of peaks passing a given point per second. These terms are related so that: $c = \nu\lambda$
- where c is the velocity of light in a vacuum.
- $1\text{nm} = 1\text{m}\mu = 10\text{Å} = 10^{-9}$ meters

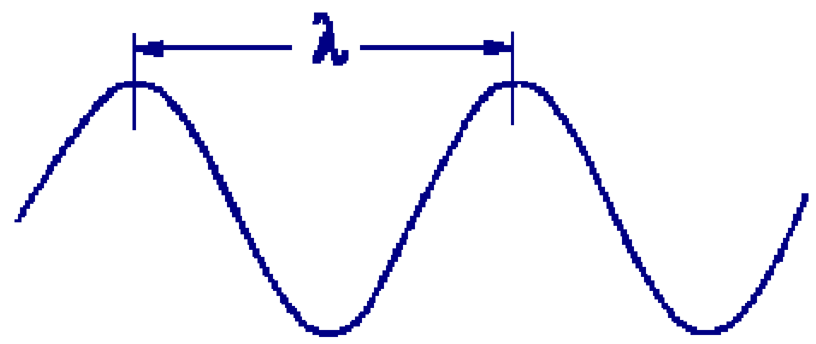
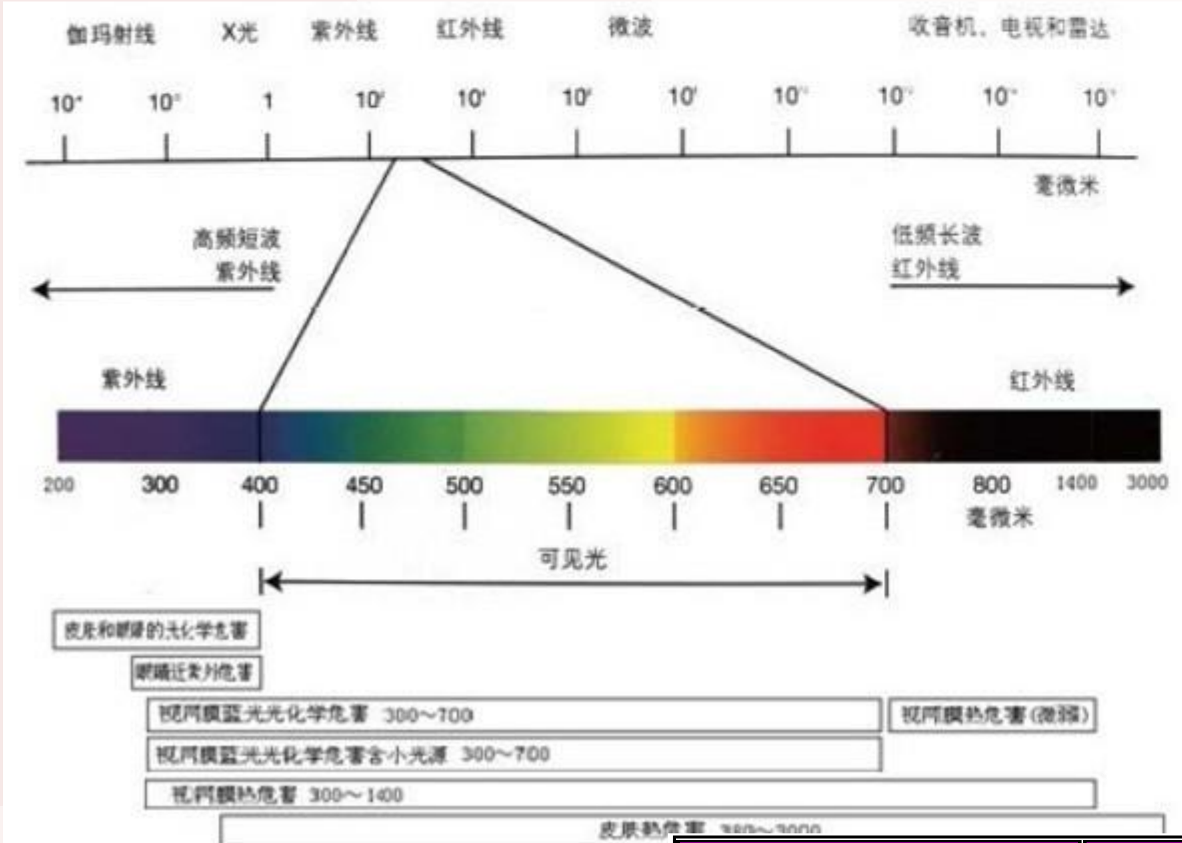
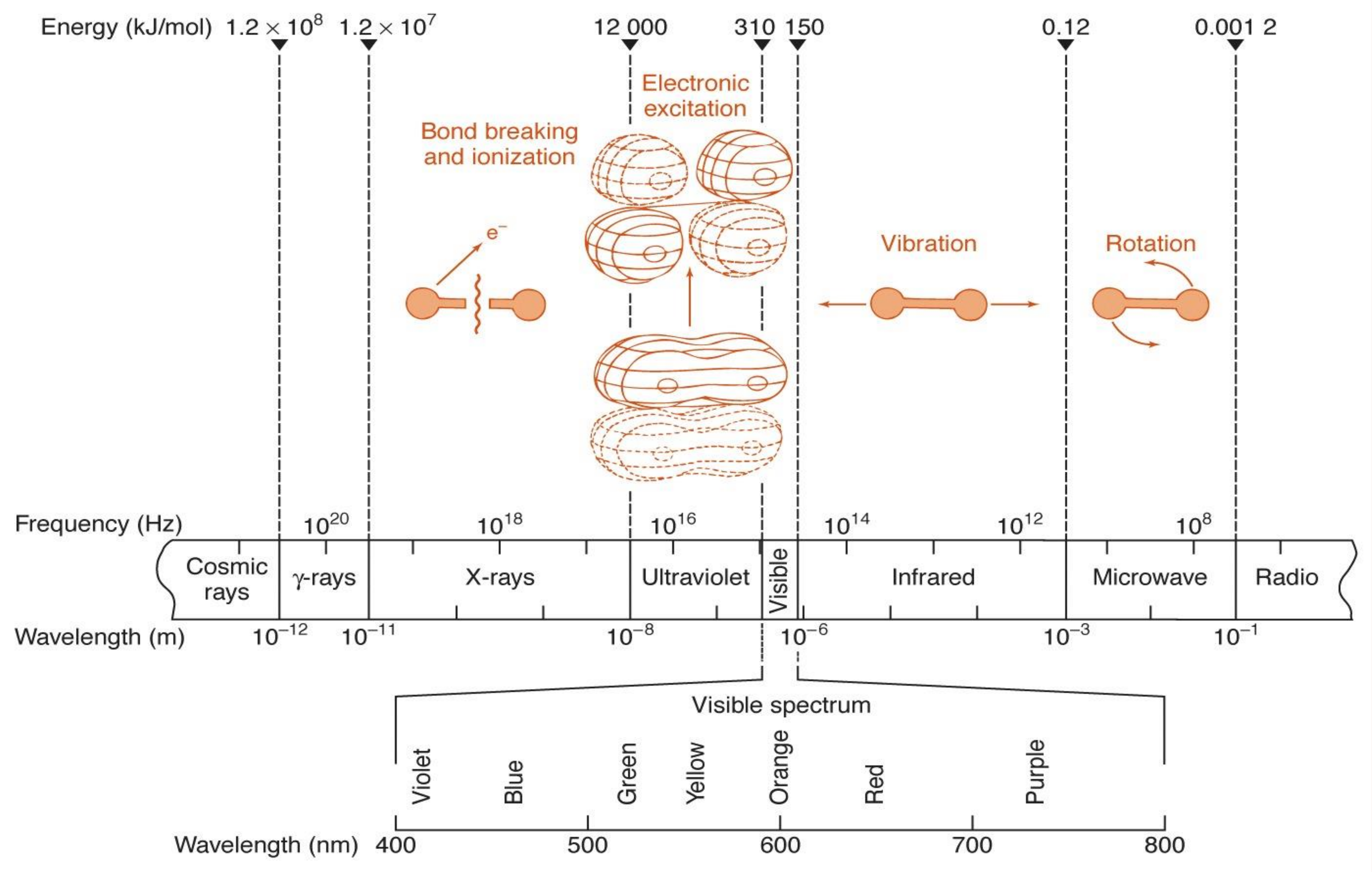


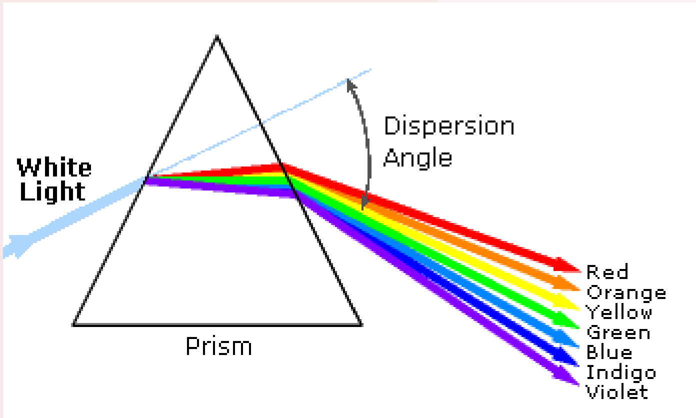
Figure 1 The wavelength λ of electromagnetic radiation

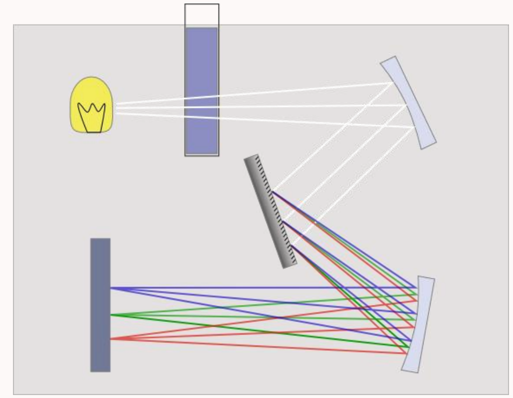


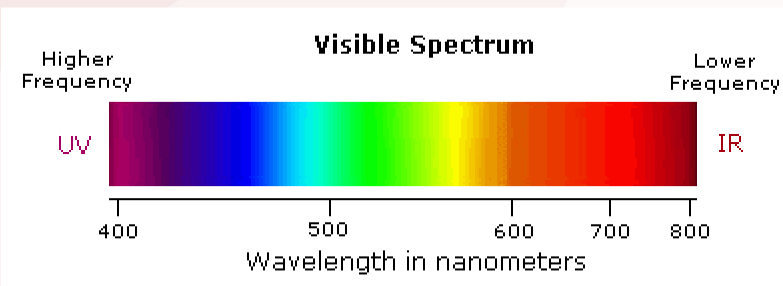
颜色	波长/nm	范围/nm
红	700	640~750
橙	620	600~640
黄	580	550~600
绿	510	480~550
蓝	470	450~480
紫	420	400~450

Region	Wavelength (nm)
Far ultraviolet	10-200
Near ultraviolet	200-380
Visible	380-780
Near infrared	780-3000
Middle infrared	3000-30,000
Far infrared	30,000-300,000
Microwave	300,000-1,000,000,000


Energy (kJ/mol) 1.2 × 10^8 1.2 × 10^7 12 000 310 150 0.12 0.001 2
Electronic excitation
Bond breaking and ionization
e⁻
Vibration
Rotation
Frequency (Hz) 10^20 10^18 10^16 10^14 10^12 10^8
Cosmic rays
γ-rays
X-rays
Ultraviolet
Visible
Infrared
Microwave
Radio
Wavelength (m) 10^−12 10^−11 10^−8 10^−6 10^−3 10^−1
Visible spectrum
Violet
Blue
Green
Yellow
Orange
Red
Purple
Wavelength (nm) 400 500 600 700 800







- **Violet:** 400 - 420 nm
- **Indigo:** 420 - 440 nm
- **Blue:** 440 - 490 nm
- **Green:** 490 - 570 nm
- **Yellow:** 570 - 585 nm
- **Orange:** 585 - 620 nm
- **Red:** 620 - 780 nm

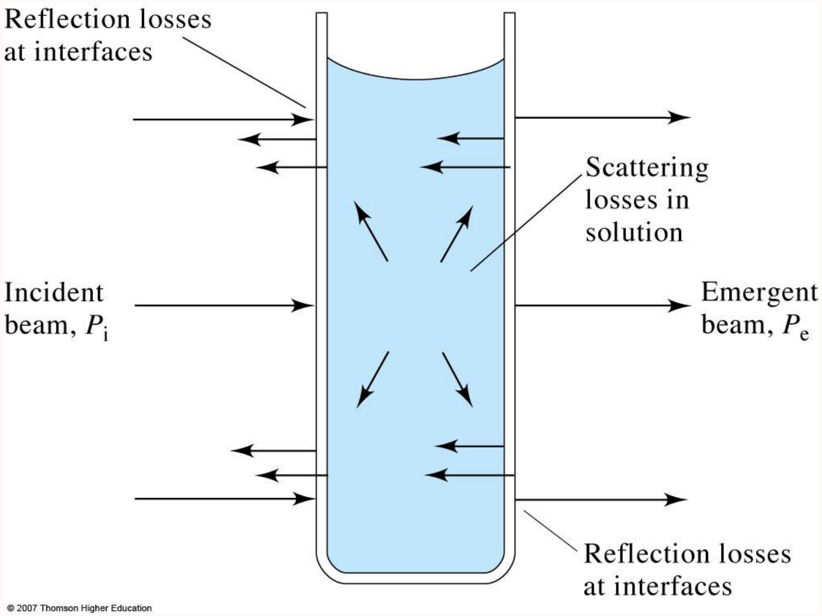
Quantum Theory

- To gain an understanding of the origins of practical absorption spectrometry, a short diversion into quantum theory is necessary.
- For this purpose, it is best to think of radiation as a stream of particles known as photons instead of the waves considered earlier. Atoms and molecules exist in a number of defined energy states or levels and a change of level requires the absorption or emission of an integral number of a unit of energy called a quantum, or in our context, a photon.
- The energy of a photon absorbed or emitted during a transition from one molecular energy level to another is given by the equation
- $e=h\nu$
- where h is known as Planck's constant (4.13566743(35)×10^(-15) eV●s) and ν is the frequency of the photon. We have already seen that $c=\nu\lambda$, therefore, $E=hc/\lambda$

the shorter the wavelength, the greater the energy of the photon and vice versa.

光的传播/吸收/衰减
Transmission and absorbance and losses

- 光强度在传播某一个特定物体时的减弱可以用来定量该物质的量
- 当一束平行的单色光通过含有均匀的吸光物质的吸收池（或气体、固体）时，光的一部分被溶液吸收，一部分透过溶液，一部分被吸收池表面反射
- The reduction in the intensity of light transmitted through a sample can be used to quantitate the amount of an unknown material.



透射比

$$T = \frac{P}{P_0} \approx \frac{P_{sample}}{P_{blank}}$$

$$A = -\log T = \log \frac{P_0}{P} \approx \log \frac{P_{blank}}{P_{sample}}$$

Lambert-Beer定律
Beer' s Law

- Quantitative relationship between absorbance and concentration of analyte
- Absorption is additive for mixtures

$$A = \log\frac{P_0}{P} = \varepsilon bc$$

ε = molar absorptivity

b = pathlength

c = concentration

- 说明物质对单色光吸收的强弱与吸光物质的浓度（c）和液层厚度 （b）间的关系的定律，是光吸收的基本定律，是紫外-可见光度法定量的基础。被测组分对紫外光或可见光具有吸收，且吸收强度与组分浓度成正比

Really: $A_\lambda = \varepsilon_\lambda bc$

Beer's Law is always wavelength-specific

$$A_{mixture} = A_1 + A_2 + \ldots + A_n$$

$$A_{mixture} = \varepsilon_1 bc_1 + \varepsilon_2 bc_2 + \ldots \varepsilon_n bc_n$$

$$A = \log \frac{P_0}{P} = \varepsilon bc$$ 吸光度

ε = molar absorptivity 摩尔吸光系数

b = pathlength 光程

c = concentration

当 c用 mol/L 、b 用 cm 为单位时，用摩尔吸光系数 e 表示，单位为 L/（mol·cm）

吸光系数：当 c 用 g /L，b 用 cm 为单位时，K 用吸光系数 a 表示，单位为 L/（g·cm）

$A = a \cdot b \cdot c$

molar mass

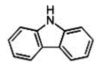
e 与 a 之间的关系为：$e = M \cdot a$

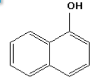
Molar absorptivity is a constant for a particular substance (at a wavelength).

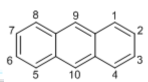
Tab. 2.1. Examples of molar absorption coefficients, ε (at the wavelength corresponding to the maximum of the absorption band of lower energy). Only approximate values are given, because the value of ε slightly depends on the solvent

Compound	ε*/L mol*$^{-1}$ *cm*$^{-1}$	*Compound*	ε*/L mol*$^{-1}$ *cm*$^{-1}$
Benzene	≈200	Acridine	≈12 000
Phenol	≈2 000	Biphenyl	≈16 000
Carbazole	≈4 200	Bianthryl	≈24 000
1-Naphthol	≈5 400	Acridine orange	≈30 000
Indole	≈5 500	Perylene	≈34 000
Fluorene	≈9 000	Eosin Y	≈90 000
Anthracene	≈10 000	Rhodamine B	≈105 000
Quinine sulfate	≈10 000		

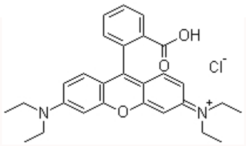
咔唑

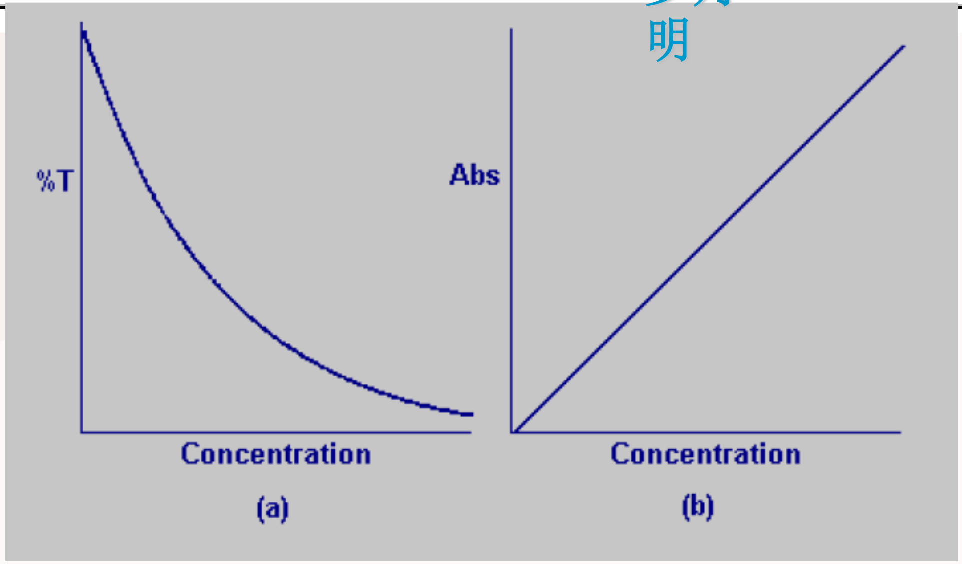






罗丹明



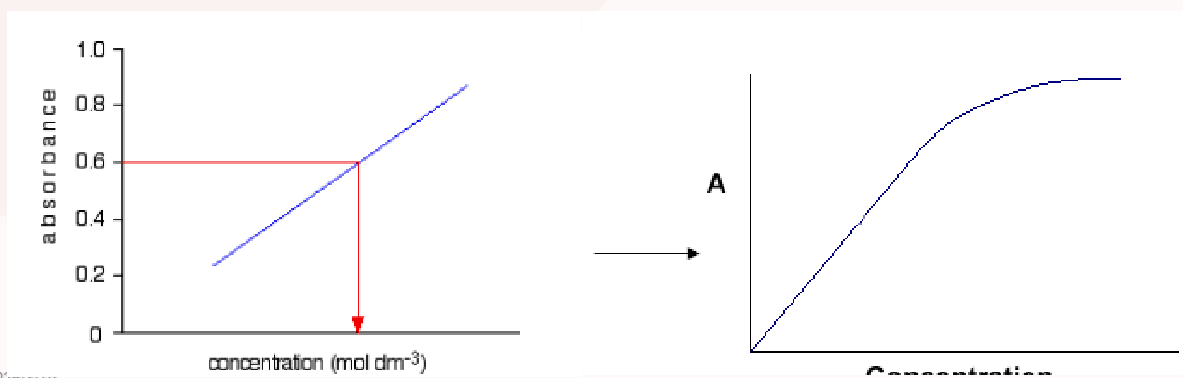


局限性

How might one avoid?

Limitations and deviations from Beer's Law

- L-B 定律是一个有限的定律，只有在稀溶液中才能成立。由于在高浓度时（通常 $C > 0.01mol/L$），吸收质点之间的平均距离缩小到一定程度，邻近质点彼此的电荷分布都会相互受到影响，此影响能改变它们对特定辐射的吸收能力，相互影响程度取决于C，因此，此现象可导致A与C线性关系发生偏差
- The Beer-Lambert Law breaks down for solutions of higher concentration.



局限性

Limitations and deviations from Beer’ s Law

1 Instrumental
2 Polychromatic radiation
3 Different molar absorptivities at different wavelength leads to non-linearities in Beer’s Law
4 Stray radiation
5 Mistmatched cells
6 Non-zero intercept in calibration curve

How to make a UV-vis absorption measurement

1) Make a 0%T (dark current) measurement

2) Make a 100%T (blank) measurement

3) Measure %T of sample

4) Determine %T ratio and thus the absorbance value

What is UV-visible absorption measuring?

紫外吸收光谱：分子价电子能级跃迁。
波长范围：100-800 nm.
(1) 远紫外光区:100-200nm
(2) 近紫外光区:200-400nm
(3)可见光区:400-800nm
可用于结构鉴定和定量分析。
电子跃迁的同时，伴随着振动转动能级的跃迁;带状光谱。

- The absorption of a photon generates an electronic excited state

- UV-vis energy often matches up with transitions of bonding electrons
  Often relatively short lifetimes (1-10 nsec)

- Relaxation can occur non-radiatively

- or by emission of radiation (fluorescence or phosphorescence)

$\mathbf{M} + h\nu \rightarrow$ M * → M + 热 / M + 荧光或磷光

基态 激发态

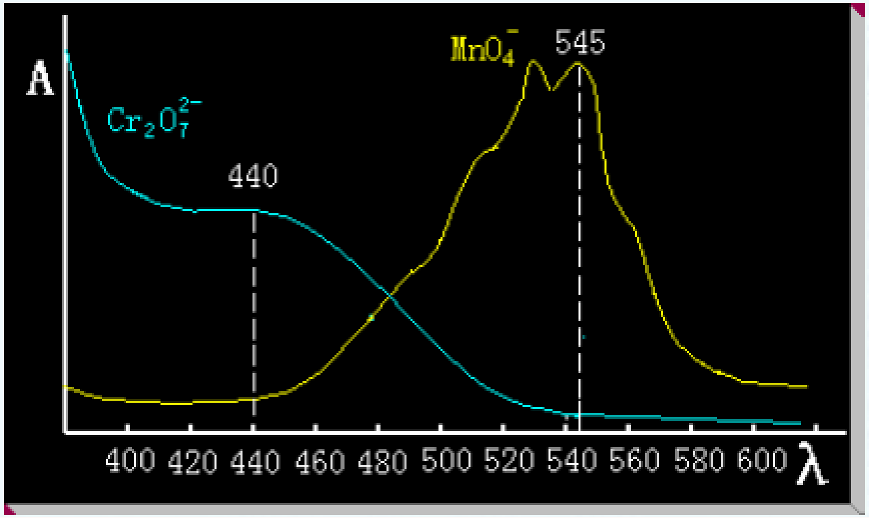
E_1 （ΔE） E_2

$\Delta E = E_2 - E_1 = h\nu$

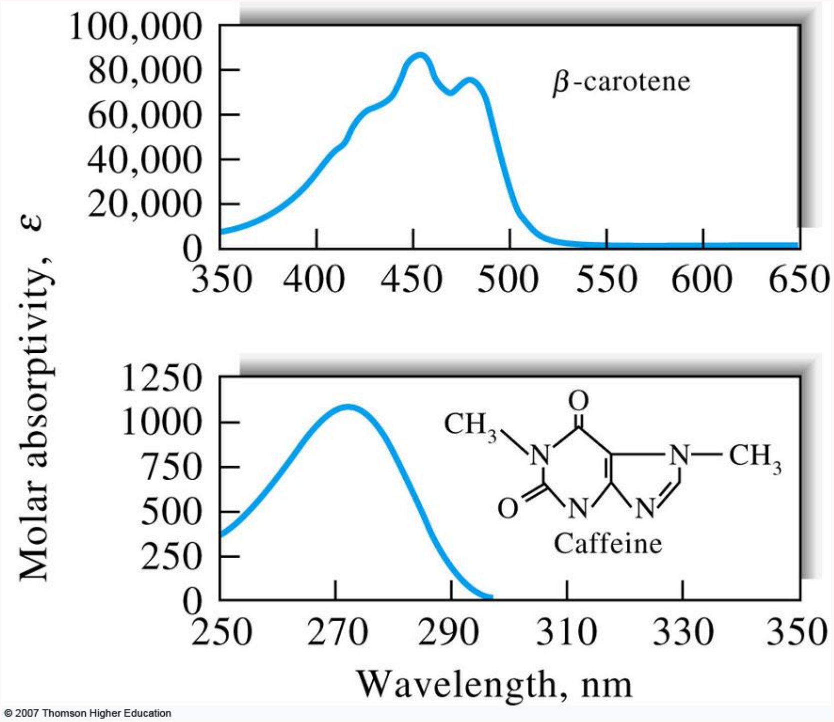
量子化；选择性吸收

吸收曲线与最大吸收波长λ max

用不同波长的单色光照射，测吸光度;



吸收曲线



①同一种物质对不同波长光的吸光度不同。吸光度最大处对应的波长称为最大吸收波长**λmax**

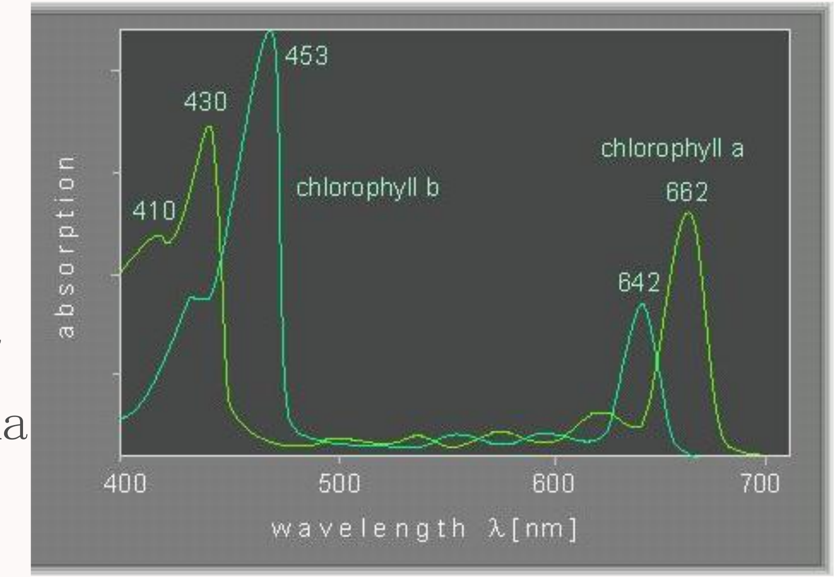
②不同浓度的同一种物质，其吸收曲线形状相似**λmax**不变。而对于不同物质，它们的吸收曲线形状和**λmax**则不同。

③吸收曲线可以提供物质的结构信息，并作为物质定性分析的依据之一。

吸收曲线= the dependence of absorbance from wavelength

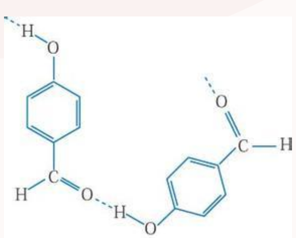
From the absorption spectrum can be determined:

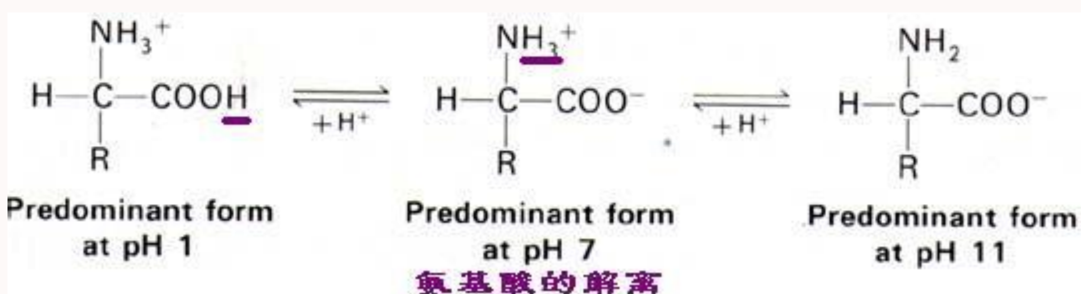
- the position of the absorption maxima
- the intensity of the absorption maxima



影响因素

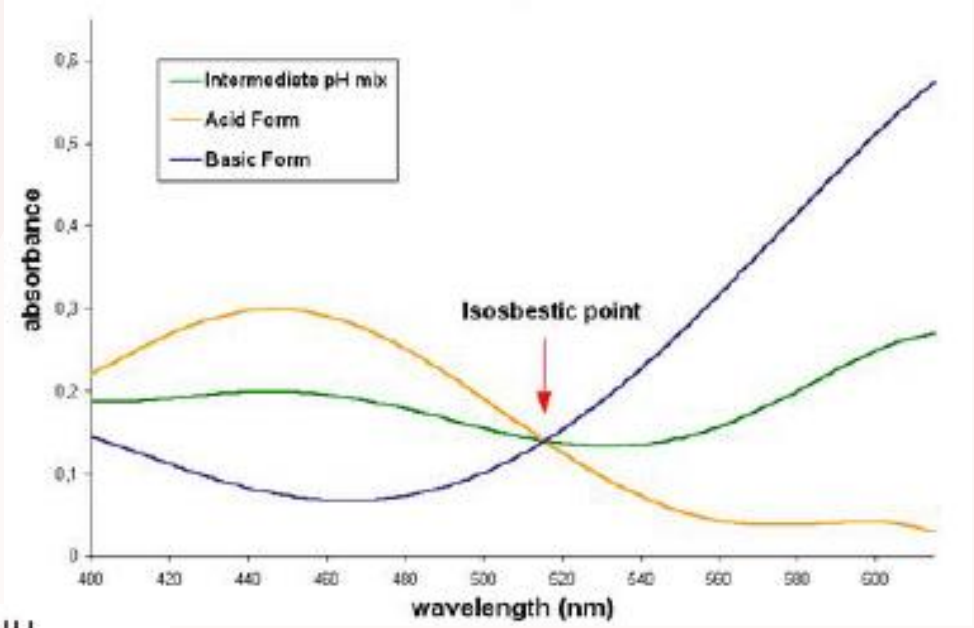
- Temperature (at higher temperatures monomers are predominant)
- Concentration (at higher concentrations dimers/polymers are predominant)
- pH (influencing the balance between ionized and non-ionized forms)



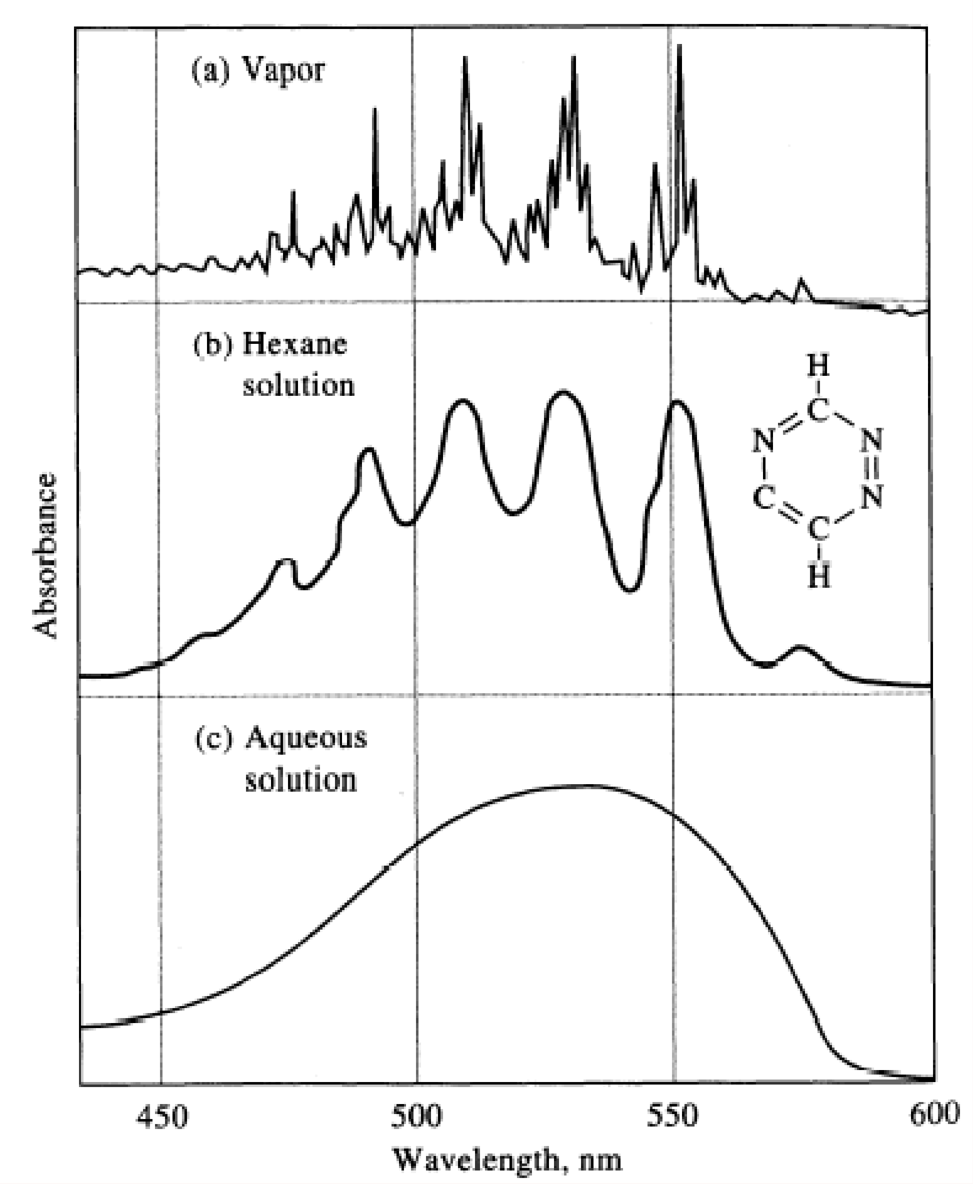


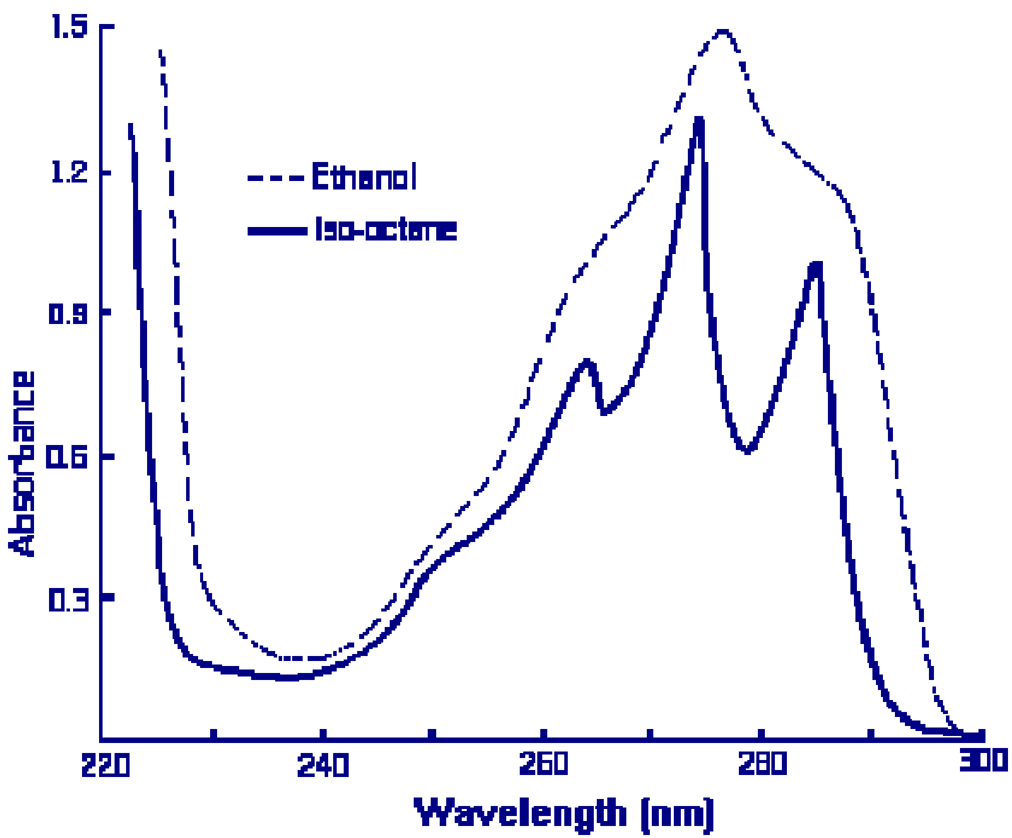
等吸收点
isosbestic point

- 浓度一定的具有光吸收性质的化合物溶液，随着溶液条件或其他影响因素(如pH、光分解、热分解等)的变化呈现不同形状的吸收曲线。在这一组吸收曲线中，可能有一个或几个共同的交点，这些交点即为等吸收点。
- izobestic point is the intersection of all absorption curves of a solution at different pH's



Solvent effects





illustrates the effect of iso-octane and ethanol on the spectrum of phenol, a change from hydrocarbon to hydroxylic solvent. The loss of fine structure in the latter is due to broad band h-bonded solvent-solute complexes replacing the fine structure present in the iso-octane. The fine structure in the latter solvent illustrates the principle that non-solvating 溶剂化 or nonchelating solvents produce a spectrum much closer to that obtained in the gaseous state.

紫外-可见光谱与结构鉴定
Absorption signatures of various organic functional groups

- Commonly observed transitions are n→π* or π→π*

  Chromophores have unsaturated functional groups

  Rotational and vibrational transitions add detail to spectra

  Single bond excitation energies (n→σ*) are in vacuum UV (λ < 185 nm) and have very low molar absorptivities

TABLE 14-1 Absorption Characteristics of Some Common Chromophores

Chromophore	Example	Solvent	λ_{max}, nm	ε_{max}	Transition Type
Alkene	$C_6H_{13}CH{=}CH_2$	*n*-Heptane	177	13,000	$\pi \rightarrow \pi^*$
Alkyne	$C_5H_{11}C{\equiv}C{-}CH_3$	*n*-Heptane	178	10,000	$\pi \rightarrow \pi^*$
			196	2000	—
			225	160	—
Carbonyl	$CH_3\underset{\underset{O}{\|}}{C}CH_3$	*n*-Hexane	186	1000	$n \rightarrow \sigma^*$
			280	16	$n \rightarrow \pi^*$
	$CH_3\underset{\underset{O}{\|}}{C}H$	*n*-Hexane	180	large	$n \rightarrow \sigma^*$
			293	12	$n \rightarrow \pi^*$
Carboxyl	CH_3COOH	Ethanol	204	41	$n \rightarrow \pi^*$
Amido	$CH_3\underset{\underset{O}{\|}}{C}NH_2$	Water	214	60	$n \rightarrow \pi^*$
Azo	$CH_3N{=}NCH_3$	Ethanol	339	5	$n \rightarrow \pi^*$
Nitro	CH_3NO_2	Isooctane	280	22	$n \rightarrow \pi^*$
Nitroso	C_4H_9NO	Ethyl ether	300	100	—
			665	20	$n \rightarrow \pi^*$
Nitrate	$C_2H_5ONO_2$	Dioxane	270	12	$n \rightarrow \pi^*$

$$\varepsilon = \frac{A}{bc}$$

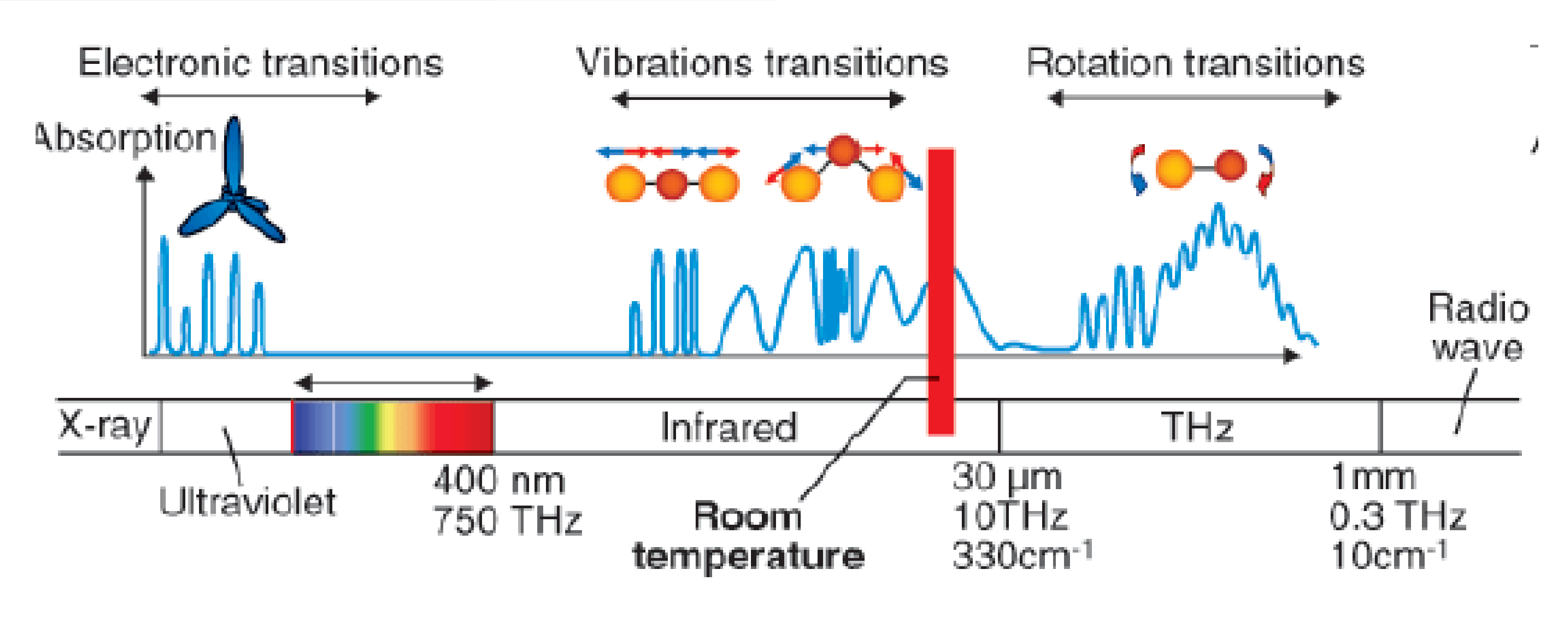
e normalized with respect to path length and concentration

电子能级和电子跃迁

- 物质分子内部三种运动形式：
- （1）电子相对于原子核的运动；
- （2）原子核在其平衡位置附近的相对振动；
- （3）分子本身绕其重心的转动。
- 分子具有三种不同能级：电子能级、振动能级和转动能级
- 三种能级都是量子化的，且各自具有相应的能量。
- 分子的内能：电子能量Ee、振动能量Ev、转动能量Er
- 即：E＝Ee+Ev+Er
- $\Delta E_e > \Delta E_v > \Delta E_r$

- （1）转动能级间的能量差 ΔE_r：0.005-0.050eV，跃迁产生吸收光谱位于远红外区。远红外光谱或分子转动光谱；
- （2）振动能级的能量差 ΔE_v 约为：0.05-1 eV，跃迁产生的吸收光谱位于红外区，红外光谱或分子振动光谱；
- （3）电子能级的能量差 ΔE_e 较大1-20eV。电子跃迁产生的吸收光谱在紫外—可见光区，紫外—可见光谱或分子的电子光谱
- （4）吸收光谱的波长分布是由产生谱带的跃迁能级间的能量差所决定，反映了分子内部能级分布状况，是物质定性的依据；
- （5）吸收谱带的强度与分子偶极矩变化、跃迁几率有关，也提供分子结构的信息。通常将在最大吸收波长处测得的摩尔吸光系数 ε_{max} 也作为定性的依据。不同物质的 λ_{max} 有时可能相同，但 ε_{max} 不一定相同；
- （6）吸收谱带强度与该物质分子吸收的光子数成正比，定量分析的依据。

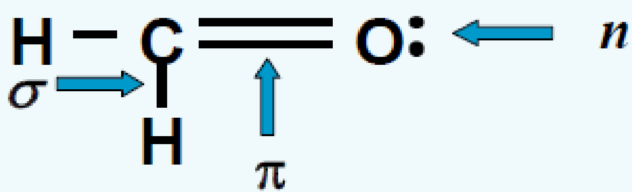
Absorption of photon results in electronic transition of a molecule.
(electrons are promoted from ground state to higher electronic states)

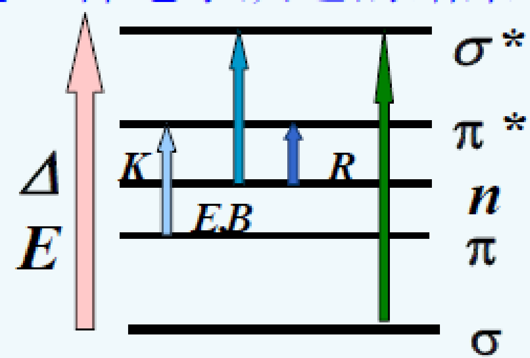


有机物吸收光谱与电子跃迁
ultraviolet spectrometry of organic compounds

1．紫外—可见吸收光谱

有机化合物的紫外—可见吸收光谱是三种电子跃迁的结果：σ 电子、π 电子、*n*电子。





分子轨道理论：成键轨道—反键轨道。

当外层电子吸收紫外或可见辐射后，就从基态向激发态（反键轨道）跃迁。主要有**四种跃迁**所需能量 Δ *E***大小顺序**为：

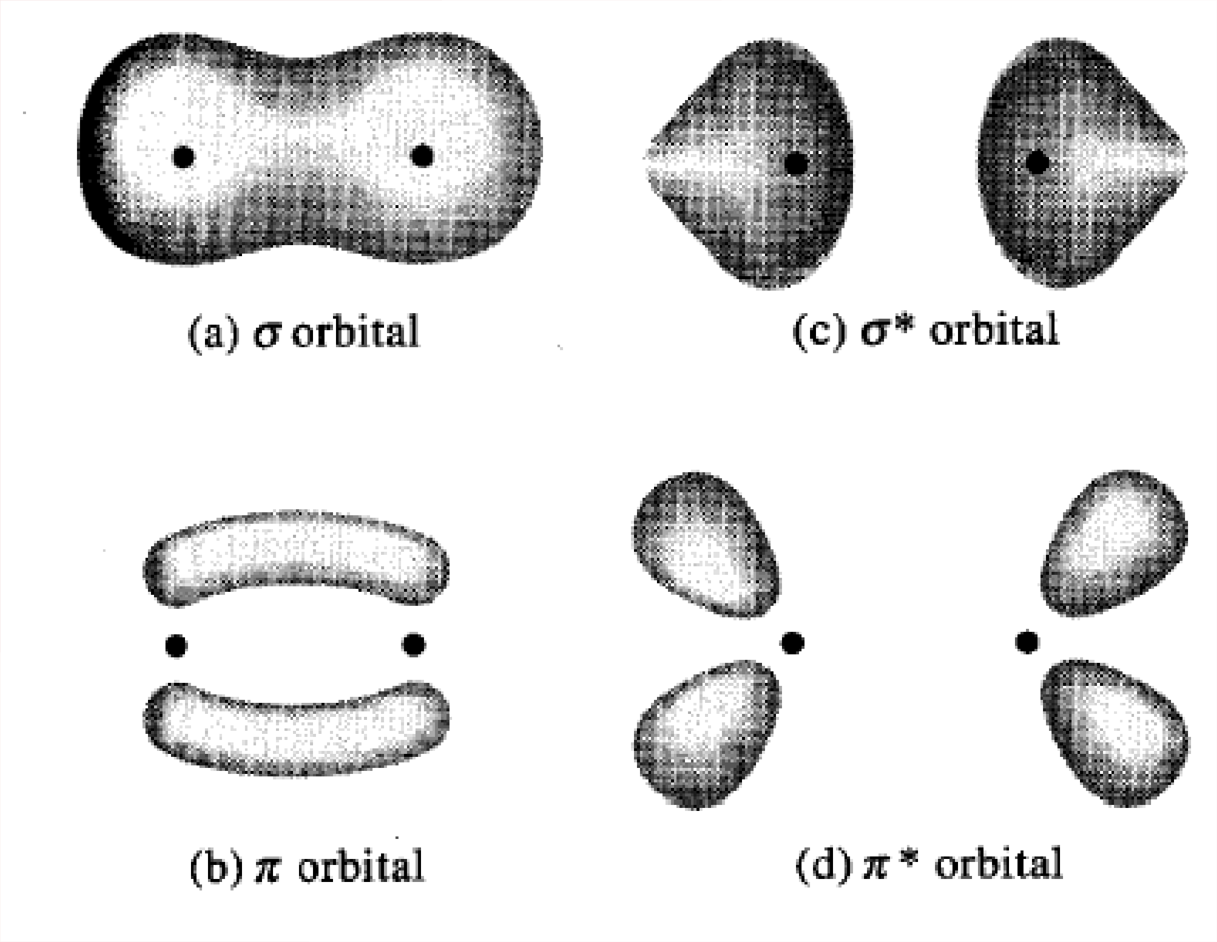
$n \rightarrow \pi^* < \pi \rightarrow \pi^* < n \rightarrow \sigma^* < \sigma \rightarrow \sigma^*$

In addition, two types of antibonding orbitals may be involved in the transition:

i) σ* (sigma star) orbital

ii) π* (pi star) orbital

(There is no such thing as an n* antibonding orbital as the n electrons do not form bonds).


(a) σ orbital
(c) σ* orbital
(b) π orbital
(d) π * orbital

Potentially, three types of ground state orbitals may be involved

2　σ → σ *跃迁

所需能量最大；σ 电子只有吸收远紫外光的能量才能发生跃迁；

饱和烷烃的分子吸收光谱出现在远紫外区；

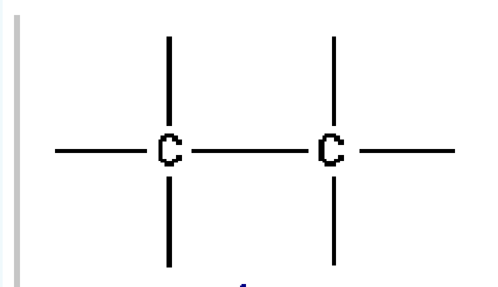
吸收波长 λ<200 nm；

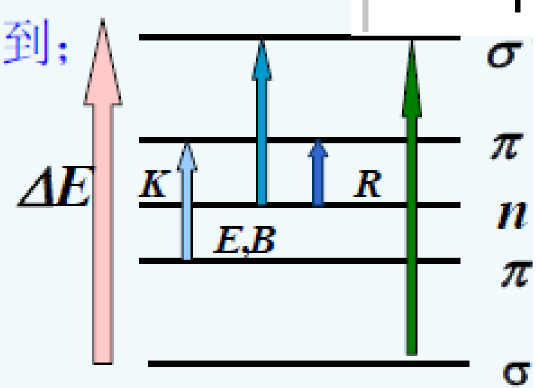
例：甲烷的 λmax为125nm ，乙烷 λmax为135nm。

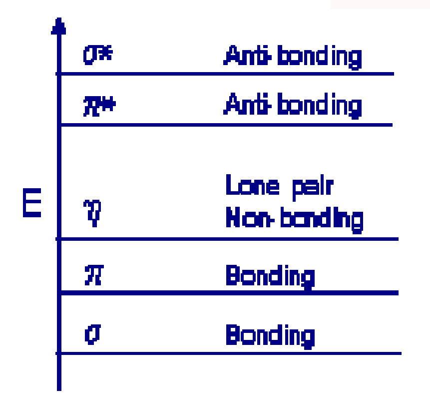
只能被真空紫外分光光度计检测到；

作为溶剂使用；

i) σ (bonding) molecular as in

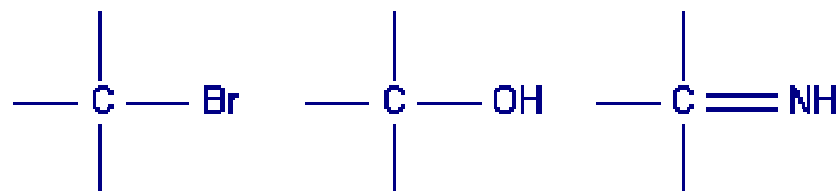






Potentially, three types of ground state orbitals may be involved

iii) n (non-bonding) atomic orbital as in



3 $n \rightarrow \sigma^*$跃迁

所需能量较大。

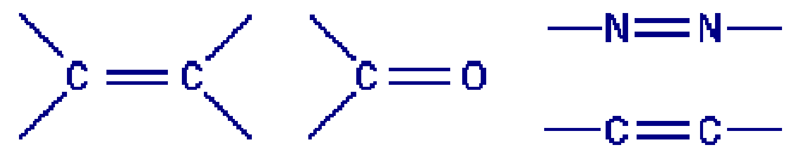
吸收波长为150～250nm，大部分在远紫外区，近紫外区仍不易观察到。

含非键电子的饱和烃衍生物(含N、O、S和卤素等杂原子)均呈现$n \rightarrow \sigma^*$跃迁。

化合物	λ_{max}（nm）	ε_{max}
H_2O	167	1480
CH_3OH	184	150
CH_3CL	173	200
CH_3I	258	365
CH_3NH_2	215	600

Potentially, three types of ground state orbitals may be involved

ii) π (bonding) molecular orbital as in



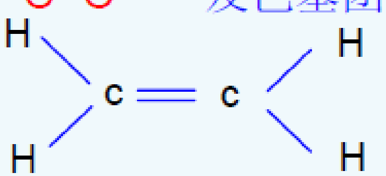
4 π → π *跃迁

所需能量较小，吸收波长处于远紫外区的近紫外端或近紫外区， ε_{max}一般在$10^4 L\cdot mol^{-1}\cdot cm^{-1}$以上，属于强吸收。

（1） 不饱和烃 π → π*跃迁

乙烯 π → π *跃迁的 λ_{max}为162nm， ε_{max}为： 1×10^4 $L\cdot mol^{-1}\cdot cm^{-1}$。 *K*带——共轭非封闭体系的π → π * 跃迁

C=C 发色基团， 但 π → π*<200nm。



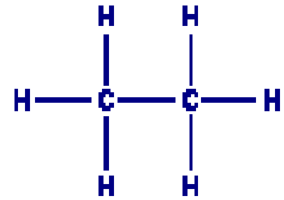
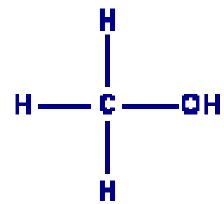
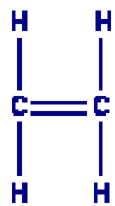
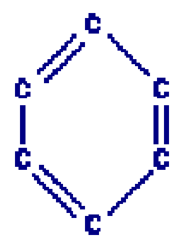
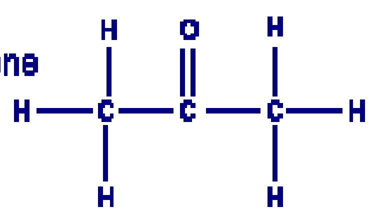
λ_{max}=162nm

助色基团取代 π → π（*K*带)发生红移。

取代基	-SR	$-NR_2$	-OR	-Cl	CH_3
红移距离	45(nm)	40(nm)	30(nm)	5(nm)	5(nm)

- Both σ to σ^* and n to σ^* transitions require a great deal of energy and therefore occur in the far ultraviolet region or weakly in the region 180-240nm. Consequently, **saturated groups** do not exhibit strong absorption in the ordinary ultraviolet region.

- Transitions of the **n to π^* and π to π^*** type occur in molecules with **unsaturated centers**; they require less energy and occur at longer.

Molecule		Transition	λ_{max} (n
Ethane		$\sigma \rightarrow \sigma^*$	135
Methanol		$\sigma \rightarrow \sigma^*$ $\eta \rightarrow \sigma^*$	150 183
Ethylene		$\pi \rightarrow \pi^*$	175
Benzene		$\pi \rightarrow \pi^*$	254
Acetone		$\eta \rightarrow \pi^*$	290

分子的共轭体系

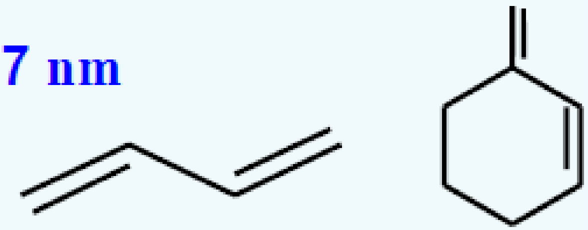
Correlation of Molecular Structure and Spectra Conjugation

- π to π * transitions, when occurring in isolated groups in a molecule, give rise to absorptions of fairly low intensity. However, conjugation of unsaturated groups in a molecule produces a remarkable effect upon the absorption spectrum. The wavelength of maximum absorption moves to a longer wavelength and the absorption intensity may often increase

共轭烯烃（不多于四个双键）$\pi \rightarrow \pi^*$跃迁吸收峰位置可由***伍德沃德——菲泽***规则估算。 $\lambda_{max} = \lambda_{基} + \Sigma n_i \lambda_i$

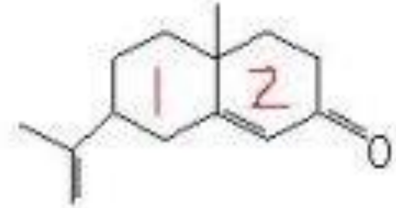
$\lambda_{基}$-----是由非环或六环共轭二烯母体决定的基准值；

无环、非稠环二烯母体： λ_{max}=217 nm

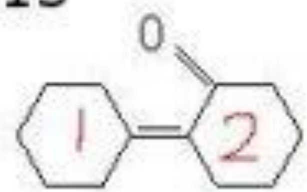


α，β-不饱和醛、酮紫外吸收计算值

计算举例



（i）六元环 α，β—不饱和酮基本值	215
2个 β 取代	+12×2
1个环外双键	+5
计算值	244nm（251nm）



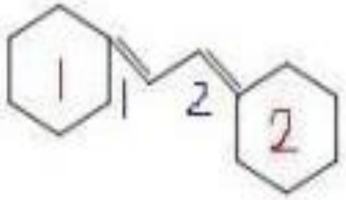
（ii）六元环 α，β—不饱和酮基本值	215
1个烷基 α 取代	+10
2个烷基 β 取代	+12×2
2个环外双键	+5×2
计算值	259nm（258nm）

(iii) 链状共轭双键　217

4个烷基取代　$+5\times4$

2个环外双键　$\underline{+5\times2}$

计算值　247nm（247nm）



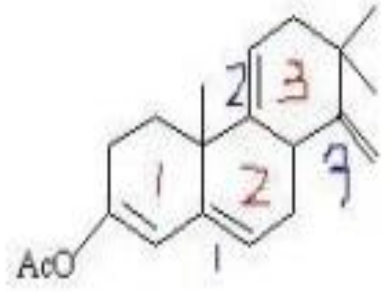
(iv) 同环共轭双烯基本值　253

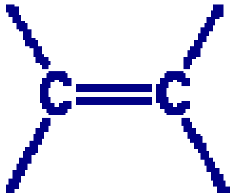
5个烷基取代　$+5\times5$

3个环外双键　$+5\times3$

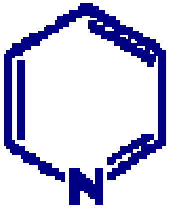
延长一个双键　$\underline{+30\times2}$

计算值　353nm（355nm）

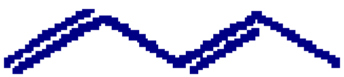




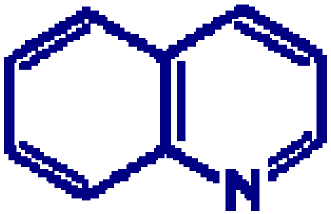
λ_{max} 180-200 nm
ε = 10,000



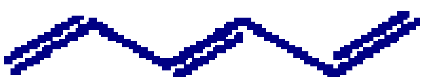
λ_{max} 250 nm



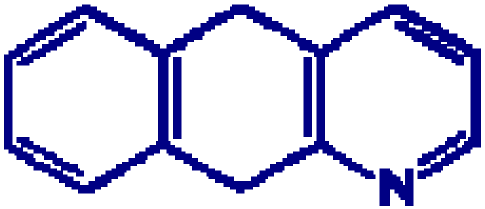
λ_{max} 217 nm
ε = 21,000



λ_{max} 290 nm



λ_{max} 258 nm
ε = 35,000



λ_{max} 360 nm

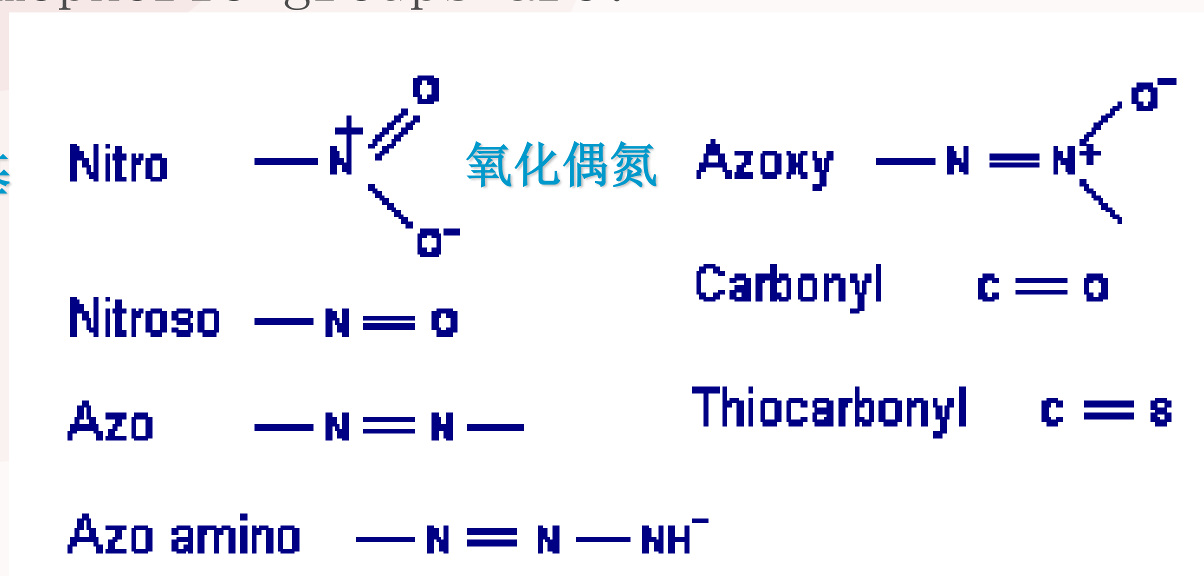
生色团Chromophores

- A chromophore (literally color-bearing) group is a functional group, not conjugated with another group, which exhibits a characteristic absorption spectrum in the ultraviolet or visible region. Some of the more important chromophoric groups are:

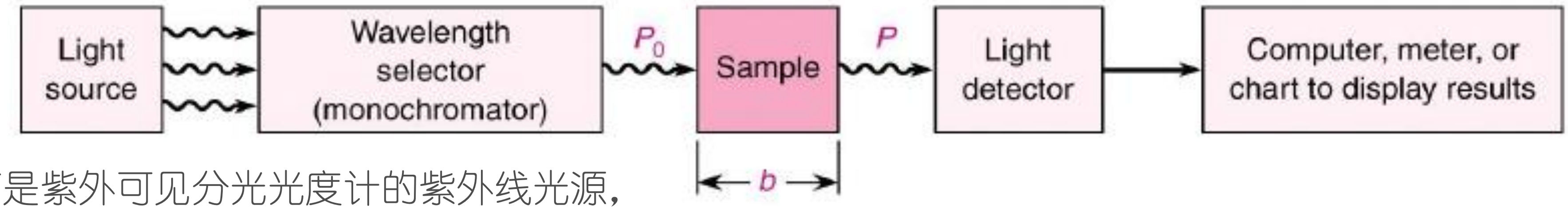
硝基

亚硝基

偶氮



Instrumentation



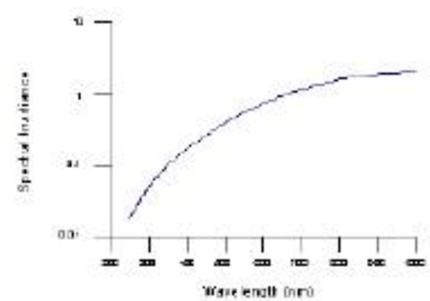
氘灯是紫外可见分光光度计的紫外线光源，
它发出的光的波长范围一般为190~400nm的连续光谱带。

Light Source

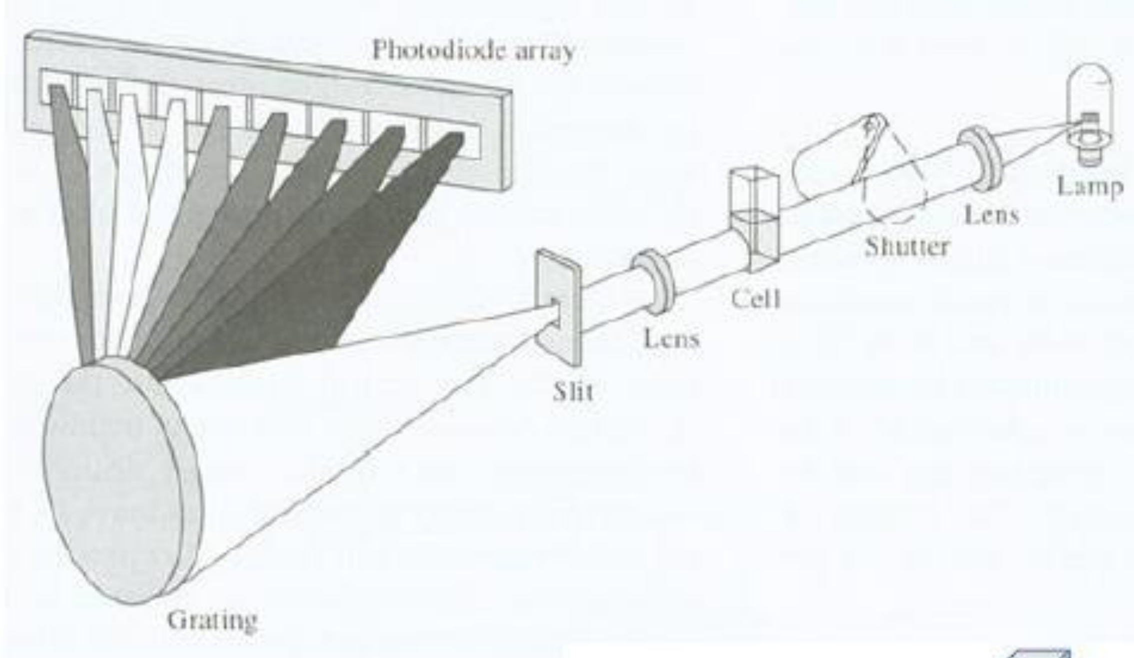
Deuterium Lamps—a truly continuous spectrum in the ultraviolet region is produced by electrical excitation of deuterium at low pressure. (160nm~375nm)

Deuterium Lamps

Tungsten Filament Lamps—the most common source of visible and near infrared radiation.



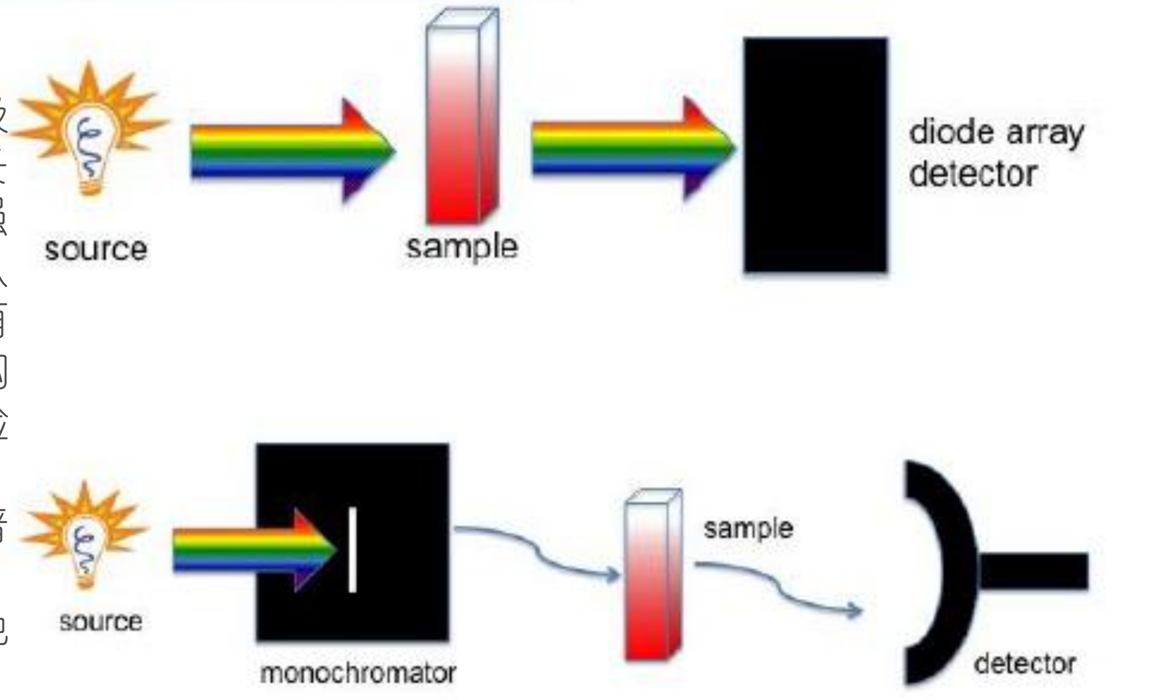
Tungsten Lamp



Photodiode array (CCD) allow simultaneous detection of all wavelength.

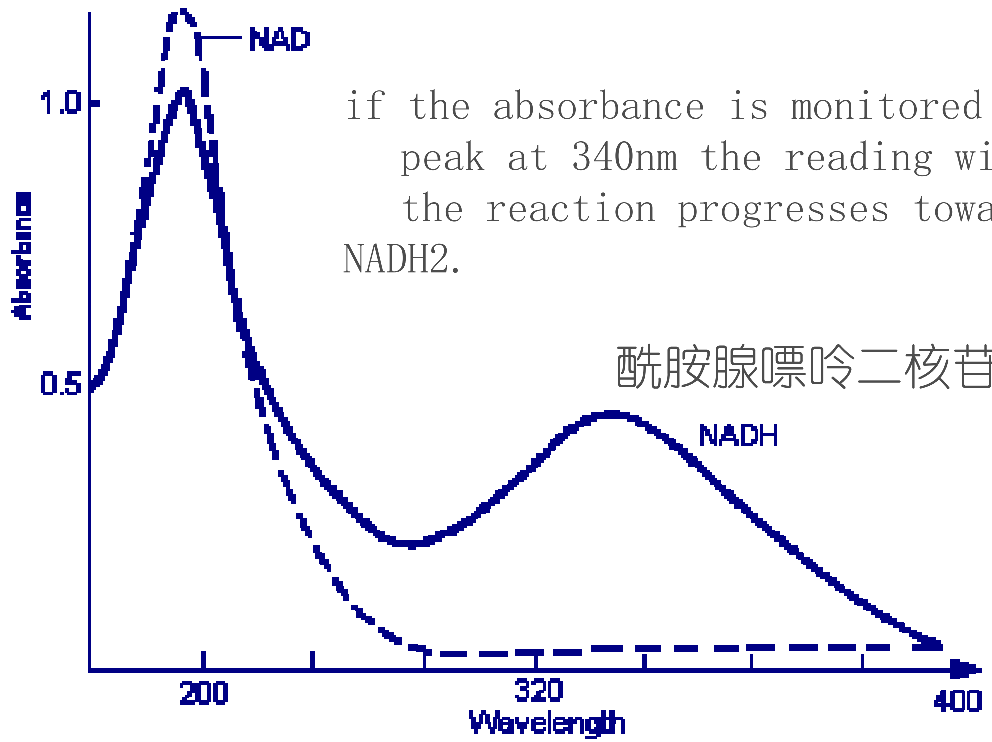
No monochromator is needed!

复色光通过样品池被组分选择性吸收后再进入单色器，照射在二极管阵列装置上，使每个纳米波长的光强度转变为相应的电信号强度，即获得组分的吸收光谱，从而获得特定组分的结构信息，有助于未知组分或复杂组分的结构确定。用一组光电二极管同时检测透过样品的所有波长紫外光，而不是某一个或几个波长，和普通的紫外-可见分光检测器不同的是进入流动池的光不再是单色光。



Applications

烟酰胺腺嘌呤二核苷酸



if the absorbance is monitored at the NADH2 peak at 340nm the reading will increase as the reaction progresses towards NADH2.

酰胺腺嘌呤二核苷酸

Figure 12 Absorption spectra of NAD and NADH

检测反应速率

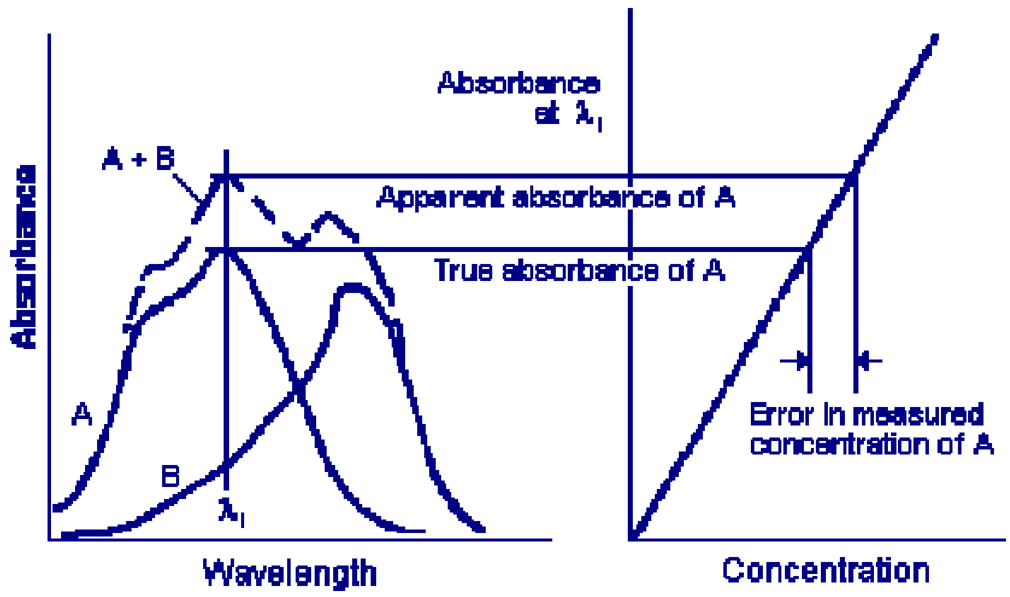


Figure 15 Error caused by superimposed absorption in a mixture

检测物质的纯度

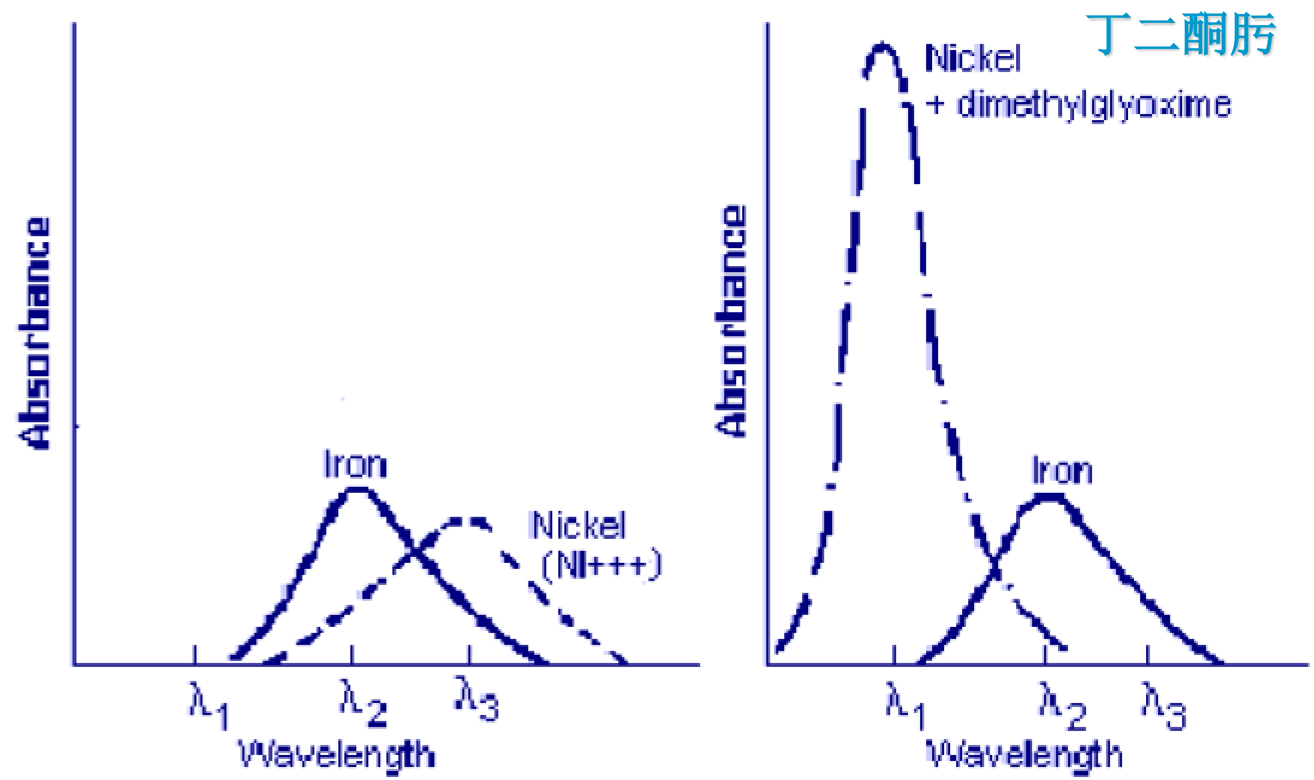


Figure 16 Improving sensitivity by adding a chemical reagent

显色法增加检测灵敏度

- Accuracy 精度This is the agreement between the value displayed on the spectrophotometer and the true value.
- Precision 精密度

Precision may be defined as the agreement (concordance) of a series of measurements of the same quantity; i.e., precision expresses the reproducibility or repeatability of a measurement.

- Reproducibility 重现性

试剂显色法

- 若分离后的化合物在紫外光或可见光下不能显示斑点，可根据被检出化合物的理化性质选择适当的显色剂使之生成颜色或荧光稳定、轮廓清楚、灵敏度高、专属性强的斑点。这种显色法是通过一种或几种试剂与被检物质产生化学反应，生成有色物质，显色法是平面色谱中广泛应用的定位方法。

显色试剂

- 显色剂可以分成两大类：一类是检查一般有机化合物的通用显色剂；另一类是根据化合物分类或特殊官能团设计的专属性显色剂。
- 通用显色剂
- （1）硫酸常用的有四种溶液：硫酸-水（1:1）溶液；硫酸-甲醇或乙醇（1:1）溶液；1.5mol/l硫酸溶液与0.5-1.5mol/l硫酸铵溶液，喷后110℃烤15min，不同有机化合物显不同颜色。
- （2）0.5%碘的氯仿溶液对很多化合物显黄棕色。
- （3）中性0.05%高锰酸钾溶液易还原性化合物在淡红背景上显黄色。
- （4）碱性高锰酸钾试剂还原性化合物在淡红色背景上显黄色。
- 溶液Ⅰ：1%高锰酸钾溶液；
- 溶液Ⅱ：5%碳酸钠溶液；

专属性显色剂

- （1）烃类
- ①硝酸银、过氧化氢
- 检出物：卤代烃类。
- （2）醇类
- ①3，5-二硝基苯酰氯
- 检出物：醇类。
- （3）有机酸类
- ①溴甲酚绿
- 检出物：有机酸类。
- 水合茚三酮作为指纹的显色剂
- 比色法测多糖-苯酚-硫酸法显色

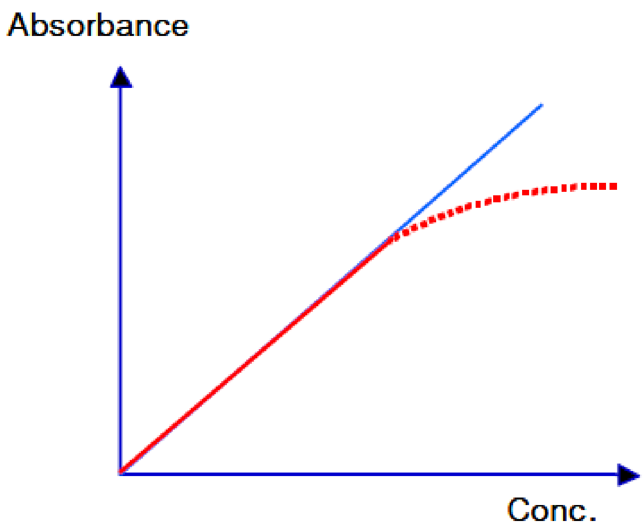
Ultraviolet-Visible Spectrophotometry

紫外—可见分光光度法

Standard Curve, Calibration curve

- Lambert-Beer's Law
  - A=abC
  - Concentration determination of Sample

- Standard Curve
  - Standard Samples
  - Proportional constant
  - Absorbance measurement of Samples



Standard Curve

Concentration determination of Sample

Structure of UV/Visible Spectrophotometer